

SAT Math

Systems of Linear Equations 3

Question #	ID		Answer
3.1	d1b66ae6	$\begin{aligned} -x + y &= -3.5 \\ x + 3y &= 9.5 \end{aligned}$ If (x, y) satisfies the system of equations above, what is the value of y ?	1.5
3.2	70feb725	During a month, Morgan ran r miles at 5 miles per hour and biked b miles at 10 miles per hour. She ran and biked a total of 200 miles that month, and she biked for twice as many hours as she ran. What is the total number of miles that Morgan biked during the month? A. 80 B. 100 C. 120 D. 160	D
3.3	e1248a5c	In the system of equations below, a and c are constants. $\frac{1}{2}x + \frac{1}{3}y = \frac{1}{6}$ $ax + y = c$ If the system of equations has an infinite number of solutions (x, y), what is the value of a ? A. $-\frac{1}{2}$ B. 0 C. $\frac{1}{2}$ D. $\frac{3}{2}$	D

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3.4	52cb8ea4	$\begin{aligned} 7x - 5y &= 4 \\ 4x - 8y &= 9 \end{aligned}$ If (x, y) is the solution to the system of equations above, what is the value of $3x + 3y$? A. -13 B. -5 C. 5 D. 13	B
3.5	d7bf55e1	A movie theater sells two types of tickets, adult tickets for \$12 and child tickets for \$8. If the theater sold 30 tickets for a total of \$300, how much, in dollars, was spent on adult tickets? (Disregard the \$ sign when gridding your answer.)	180
3.6	f718c9cf	$\begin{aligned} 5x + 14y &= 45 \\ 10x + 7y &= 27 \end{aligned}$ The solution to the given system of equations is (x, y). What is the value of xy?	1.8
3.7	466b87e3	$\begin{aligned} y &= \frac{1}{2}x + 8 \\ y &= cx + 10 \end{aligned}$ In the system of equations above, c is a constant. If the system has no solution, what is the value of c?	0.5

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3.8 e2e3942f

$$\begin{aligned} y &= 2x + 1 \\ y &= ax - 8 \end{aligned}$$

In the system of equations above, a is a constant. If the system of equations has no solution, what is the value of a ?

- A. $-\frac{1}{2}$
- B. 0
- C. 1
- D. 2

Answer

D

3.9 1e11190a

Store A sells raspberries for \$5.50 per pint and blackberries for \$3.00 per pint. Store B sells raspberries for \$6.50 per pint and blackberries for \$8.00 per pint. A certain purchase of raspberries and blackberries would cost \$37.00 at Store A or \$66.00 at Store B. How many pints of blackberries are in this purchase?

- A. 4
- B. 5
- C. 8
- D. 12

B

3.10 567ac7ab

One of the two equations in a linear system is $2x + 6y = 10$. The system has no solution. Which of the following could be the other equation in the system?

- A. $x + 3y = 5$
- B. $x + 3y = -20$
- C. $6x - 2y = 0$
- D. $6x + 2y = 10$

B

3.11 a71b1bc1

A piece of wire with a length of 32 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 4 more than 3 times the value of y . What is the value of x ?

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Question #	ID		Answer
3.12	1b1deebe	$ax + by = 72$ $6x + 2by = 56$ In the given system of equations, a and b are constants. The graphs of these equations in the xy-plane intersect at the point $(4, y)$. What is the value of a? A. 3 B. 4 C. 6 D. 14	D
3.13	ff501705	$\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$ $\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$ In the given system of equations, p is a constant. If the system has no solution, what is the value of p?	6
3.14	f75bd744	$4x - 6y = 10y + 2$ $ty = \frac{1}{2} + 2x$ In the given system of equations, t is a constant. If the system has no solution, what is the value of t?	8
3.15	adb0c96c	$24x + y = 48$ $6x + y = 72$ The solution to the given system of equations is (x, y). What is the value of y?	80
3.16	fb5e7f59	$-x - wy = -337$ $2x - wy = 47$ In the given system of equations, w is a constant. In the xy-plane, the graphs of these equations intersect at the point $(q, 19)$, where q is a constant. What is the value of w?	11

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3.17 1e0a46e4

D

Which system of linear equations has no solution?

A. $-2x + 3y = -9$
 $2x - 3y = 9$

B. $2x - 3y = 9$
 $3x + 4y = 10$

C. $2x - 3y = 9$
 $-6x + 9y = -27$

D. $-2x + 3y = 9$
 $4x - 6y = 18$

3.18 f03465dc

A

$8x + 7y = 9$
 $24x + 21y = 27$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

A. $(r, -\frac{8r}{7} + \frac{9}{7})$

B. $(-\frac{8r}{7} + \frac{9}{7}, r)$

C. $(-\frac{8r}{7} + 9, \frac{8r}{7} + 27)$

D. $(\frac{r}{3} + 9, -\frac{r}{3} + 27)$